
Design and VR/AR-based Testing of Advanced Mechatronic Systems

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Abstract

Advanced mechatronic systems with inherent partial intelligence, so-called self-optimizing systems, react autonomously and flexibly on changing environmental conditions. Such systems are capable of learning and optimizing their behavior during operation. Their principle solution represents a significant milestone because it is the result of the conceptual design as well as the basis for the concretization of the system itself, which involves experts from several domains, such as mechanics, electrical engineering/electronics, control engineering and software engineering. Today, there is no established design methodology for the design of advanced mechatronic systems. This contribution presents a new specification technique for the conceptual design of advanced mechatronic systems along with a new approach to manage the development process of such systems. We use railway technology as a complex example to demonstrate, how to use this specification technique and to what extent it facilitates the development of future mechanical engineering systems. Based on selected virtual prototypes and test beds of the RailCab we demonstrate, how VR- and AR-based approaches for a visual analysis facilitate a targeted testing of the prototypes.

Keywords

Mechatronics, Self-Optimization, Design Methodology, Principle Solution, Targeted Testing, Virtual Prototype, Visual Analysis, Virtual / Augmented Reality

1 Virtual Prototyping in the Product Innovation Process

Products and manufacturing systems of mechanical engineering and its related industrial sectors like automotive engineering are getting more and more complex. Time-to-

market is decreasing simultaneously. Under these circumstances the product innovation process is facing extraordinary challenges. Before we point out how to overcome these challenges, let us spend a brief look on the product innovation process.

The product innovation process starts from the idea of a product or business and leads to the successful product launch. It incorporates the areas of product planning, R&D and manufacturing process planning. The general work flow is shown in the figure. In practice, the product innovation process is iterative and comprises a number of cycles (see Fig. 1).

The first cycle characterizes the steps from finding the success potentials of the future to creating the promising product design, what we call the principle solution. There are four major tasks in this cycle:

- foresight
- product discovering
- business planning
- conceptual design

The aim of **foresight** is to recognize the potentials for future success, as well as the relevant business options. We use methods such as the scenario technique, Delphi studies and trend analysis.

The objective of **product discovering** is to find new product ideas. We apply in this phase creativity techniques such as the Lateral Thinking of de Bono or the well-known TRIZ.

Business planning is the final task in the cycle of strategic product planning. It initially deals with the business strategy, i.e. answering the question as to which market segments should be covered, when and how. The product strategy is then elaborated on this basis. This contains information:

- on setting out the product program
- on cost-effectively handling the large number of variants required by the market
- on the technologies used and
- on updating the program throughout the product lifecycle

Additionally, a business plan must be worked out to make sure an attractive return on investment can be achieved.

This first cycle is also concerned with the **conceptual design**, although this area of activity is actually assigned to product development in the strict sense. The result of the conceptual design is the principle solution. It is, for example, required to determine the manufacturing costs needed in the business plan. That is the reason why there is a close interaction between strategic product planning and product design linked by conceptual design. Conceptual design is the starting point for the next cycle.

This second cycle corresponds to the established understanding of product development. The essential point here is the refinement of the cross-domain principle solution by the domain experts involved, such as mechanical engineering, control technology, electronics and software engineering. The results elaborated by the domains in this cycle must be integrated into an encompassing product specification. This specification has to be verified in the light of the requirements given by the first cycle.

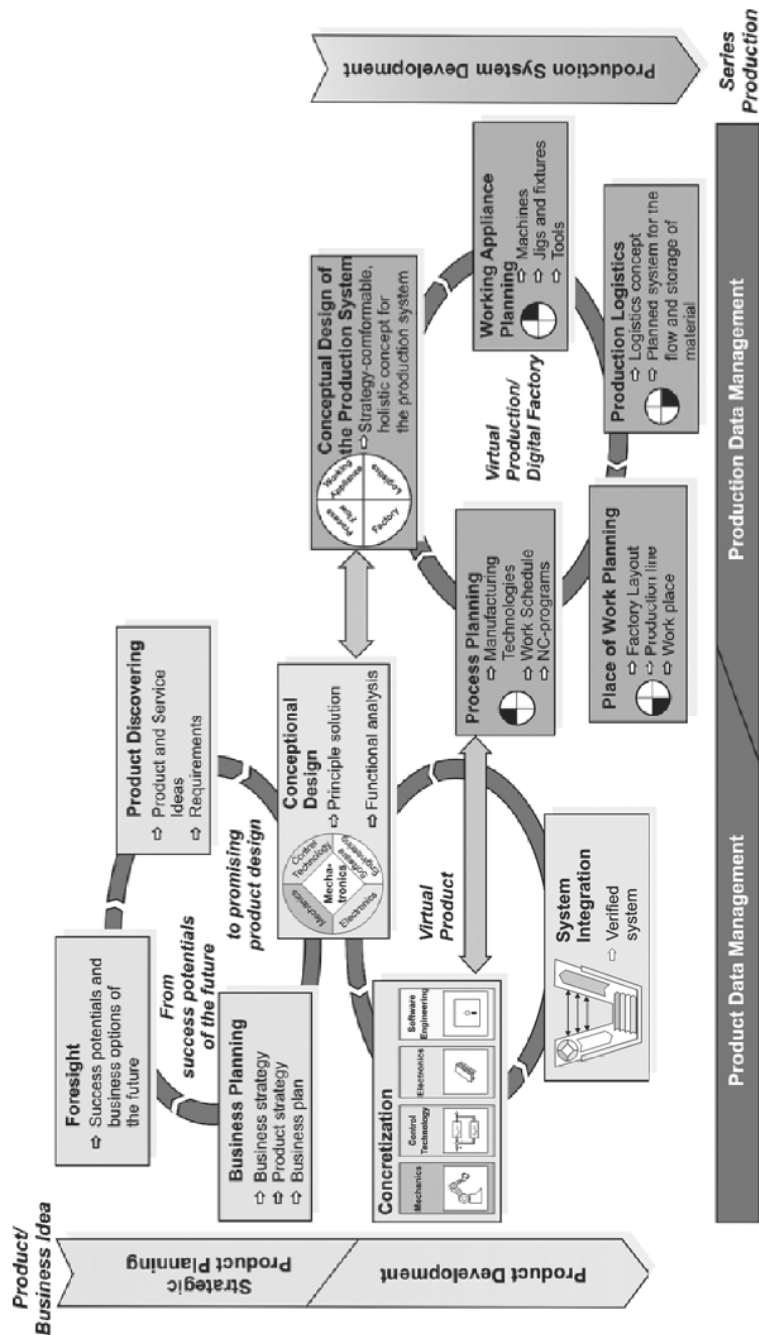


Fig. 1 The product development process as a sequence of cycles [1]

This is done in the product integration phase.

The third and last cycle focuses on manufacturing process development and the optimization of the product design with respect to manufacturing.

The second and the third cycle Product Development and Development of the corresponding manufacturing process are decisively driven by information technology. A key element is Virtual Prototyping. It means to build and analyze computer models of products and production systems being developed in order to reduce time- and cost-intensive manufacturing and testing of prototypes to a minimum. Simulation is another term for experimenting with such computer models. When we model products in the computer, we talk about the virtual product; in analogy we use the buzzwords virtual production or digital factory when we model the manufacturing system.

A perfect virtual prototype represents all aspects of a product (see Fig. 2). 3D-CAD systems are basically used to model the shape of parts. The breakdown of the product to its parts and assemblies is represented by the product structure. Therefore, it is necessary to set up a Product Data Management (PDM). The shape of individual parts in conjunction with product structure is used to develop a shape-based design of the product, what we call Digital Mock Up (DMU). It represents the spatial composition of all parts and assemblies of the product. A DMU can be used to carry out experiments such as clash detection, checking assembly and disassembly sequences. This is all based on the shape of the technical system. To analyze the behavior, we need to consider additional aspects offered by a virtual prototype.

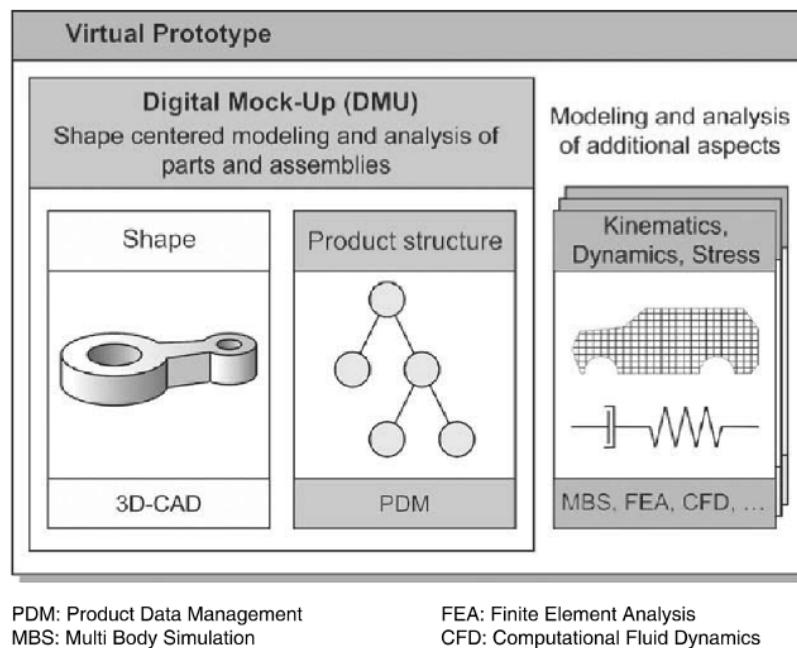


Fig. 2 From solid modeling to virtual prototyping [2]

As Fig. 2 illustrates, we consider a virtual prototype as an extension to DMU since it covers additional aspects such as kinematics, dynamics, and stress. A virtual prototype represents not only shape but also functional features and behaviors. Although, virtual prototyping cannot completely replace experiments with real prototypes, it scientifically contributes to a shorter time-to-market and less development costs even for more complex products and production systems.

2 Virtual Reality (VR) and Augmented Reality (AR)

Virtual Reality (VR) and Augmented Reality (AR) are key technologies of Virtual Prototyping. They are easy-to-understand user interfaces to a virtual design space and facilitate an interactive exploration of the functionality of a new product.

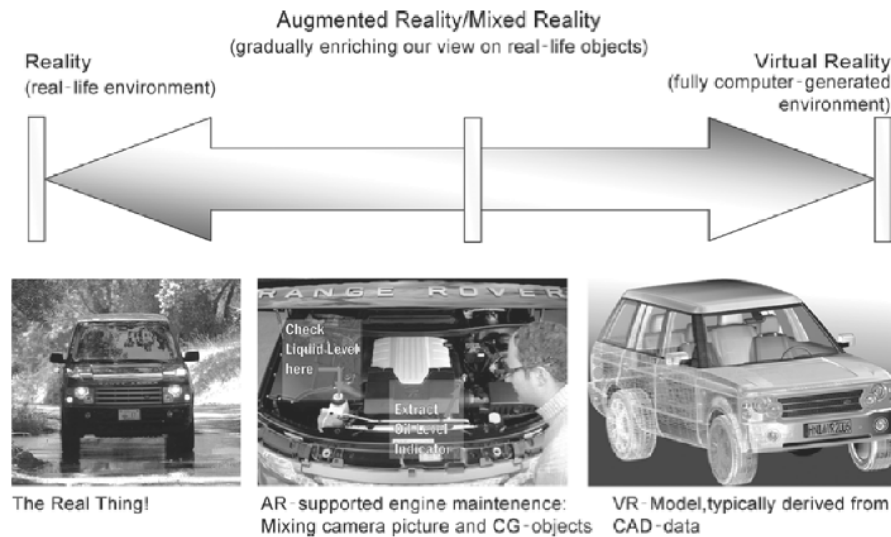


Fig. 3 From real-life to virtual reality

VR means a fully computer generated, three-dimensional environment, in which the engineer can interact with and manipulate a realistic representation of the product in real time. AR goes one step beyond: in contrast to VR, AR enriches the user's view on the real world with virtual objects, which are placed at right time and position regarding the user's perspective. Figure 3 shows an example for the transition from Reality (the real Range Rover) to AR (mixing the real engine with computer generated object for maintenance purposes) to VR (a realistic 3D-modell of the car). In details, VR technology can be characterized by the following main aspects:

Firstly, VR stands for a **realistic rendering** of the product appearance (material, surface, colors) and behavior. Secondly, VR makes use of **advanced display**

technologies that allow the engineers to experience the virtual prototype like a real one. Figure 4 (top left) shows some typical VR-display systems: Head mounted displays (HMD) with small LCD-monitors in front of the eyes. They allow a spatial view on the virtual prototype, but offer less image quality and wearing comfort.

In consequence, today most industrial VR applications use projection-based display systems that consist of several projections in different configurations. The typical configurations of such systems include PowerWall (Fig. 4, lower half) for group presentation or CAVE (Fig. 4, top right) for more spatial immersion of the users.

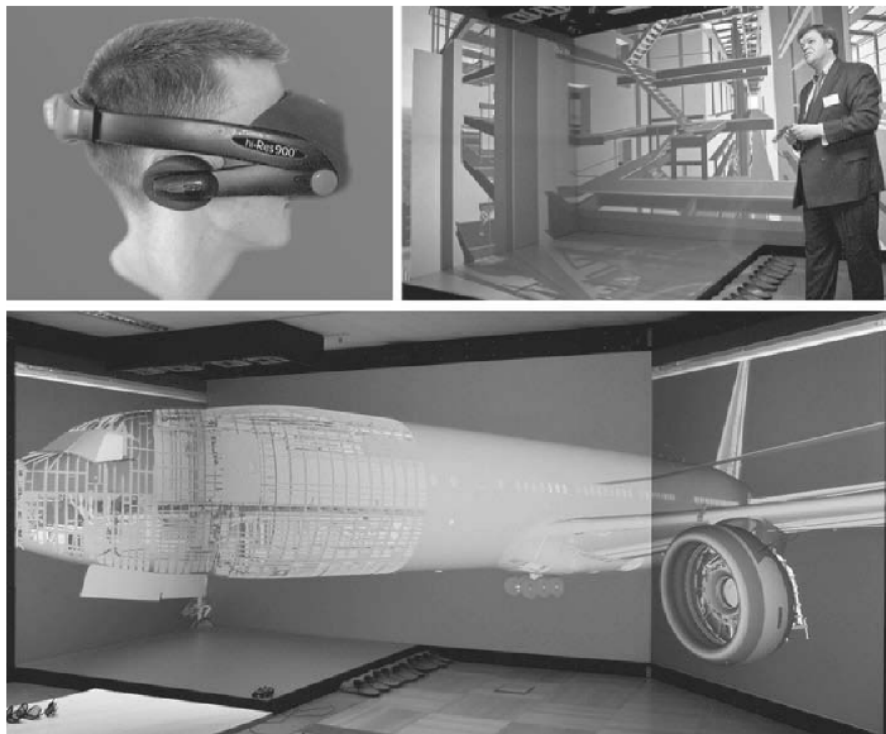


Fig. 4 Sample display devices used for VR applications: HiRes900 HMD (top left, source: Daeyang), CAVE application (top right, source: HD-Visualisation Center, HNI) and powerwall system at HNI HD-Visualisation Center (lower half, source for visualized dataset: Boeing)

In VR, the engineer has to navigate in 3D-space and to manipulate 3D-objects. Therefore, **VR-specific devices for spatial interaction** like 3D-Mouse, 3D-Wands or gloves are needed. By the help of 3D-position tracking systems, the VR system knows the position and orientation of the user in the virtual environment and is able to interpret the navigation and manipulation commands.

The main challenge of AR technology is the **context-sensitive mixture of real world elements and computer generated objects in the user's field of view**. Therefore, an exact position tracking of the user inside the real world is needed in real-

time. Based on this information, the AR-system determines the virtual objects to be shown, their size and position in the user's field of view.

Today there are two different approaches to display the mixed image of real and virtual objects: "Video-see-through" display devices have integrated miniature video cameras. The user sees a real-time video stream of his real environment, which is enriched with computer-generated objects. Figure 5 (top right) shows a video-see-through HMD published by Canon in 2002 and its application in car door assembly (see Fig. 5, top left).



Fig. 5 Sample display devices and AR applications: AR-system for vehicle assembly (top left, source: Brose), video-see-through HMD (top right, source: Canon), automotive head-up display (lower left, source: AUDI) and optical-see-through glasses (lower right, source: Zeiss)

In Fig. 5 (lower left), we see an example of "optical-see-through" display in a car. The computer generated objects are directly projected in the front window. Here, the complex and time-consuming video processing is not necessary. Figure 5 (lower right) shows a high resolution optical-see-through HMD from Zeiss.

3 Up-to-date Applications of VR and AR in Industry

In recent years, VR technology has successfully made its way from research institutes into industrial practice. VR today is a key technology in industry, e.g. in product development, plant engineering and service. It facilitates the engineer's understanding of complex design concepts and allows more efficient interaction between the engineer

and the computer. This saves time and money and finally enhances product quality. Augmented Reality is still at the beginning of its industrial employment, however first joint research projects with partners from industry show the great benefit of this fascinating technology.

The following projects of the Heinz Nixdorf Institut give a survey about how VR and AR technology could be applied in the product innovation process.

3.1 Composing Mechatronic Systems in VR

The complexity of modern mechatronic systems and the necessity to efficiently analyze and explore their large number of potential configurations and behavior patterns ask for new development methods and tools. We developed a virtual prototyping environment¹, which allows the engineers to interactively compose mechatronic prototypes in the virtual world [3,4]. The design approach we used is based on the combined effects of interconnected system elements, also referred to as “solution elements”, like sensors, actuators, and mechanical parts as well as Mechatronic Function Modules (MFM).

During the design in the virtual environment, the system composition is executed in a synthesis and analysis cycle (Fig. 6). In the first step, the system synthesis, the engineer interactively arranges the assembly structure. Therefore, he uses 3D-models of solution elements from a construction library. In parallel, the active structure of the assembled system will be automatically deduced by the design system. The active

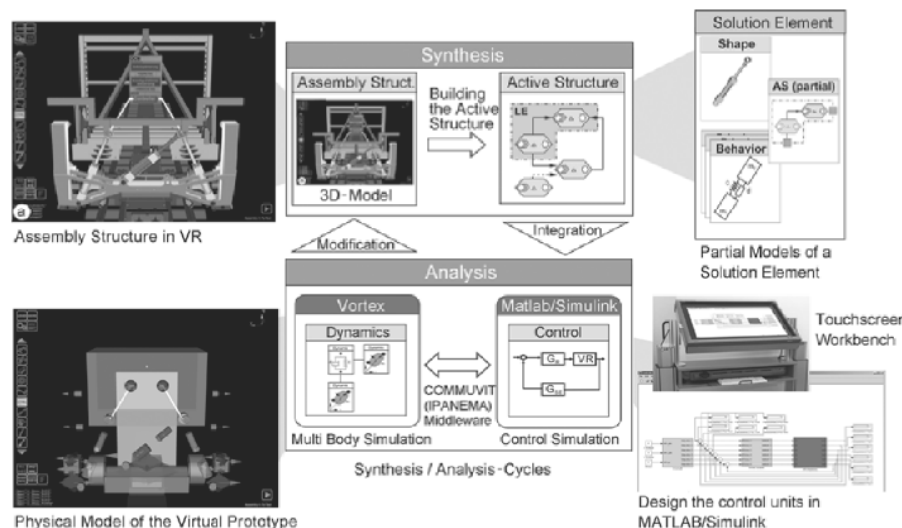


Fig. 6 Synthesis / analysis cycle in the composition of mechatronic systems

¹ This research was conducted as a part of the Collaborative Research Center 614 “Self-Optimizing Concepts and Structures in Mechanical Engineering”, which is supported by German Research Foundation.

structure comprises all relevant configurations of solution elements (controller, sensors, and actuators) and their interrelationships. In the second step, the engineer can analyse the dynamic behaviour of the designed system in real-time. During system analysis Matlab/Simulink is used for the simulation of the system control. The simulation of the multi body system is done by VORTEX.

In order to make the interaction of system composition more intuitive, a set of gestures are defined as commands for the engineer to execute the assembly actions in VR (Fig. 7).

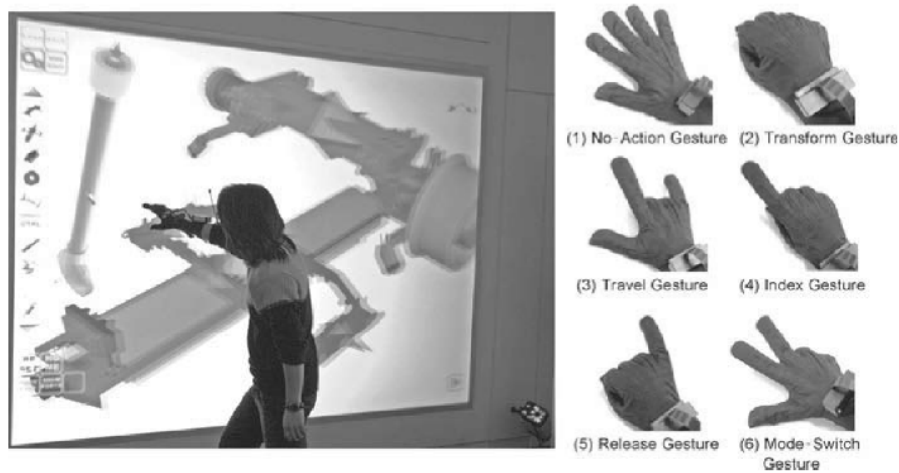


Fig. 7 Composing a virtual mechatronic prototype via intuitive gestures

To facilitate an intuitive exploration of the virtual prototype, various visual effects (see Fig. 8) were applied in the working environment. For example, animations are used to show dynamics behavior; 3D-annotations, like the strengths and directions of forces, present the simulation results more comprehensively and are easier to be observed in the virtual environment. The system enables the engineer to get a first understanding of the behavior of the mechatronic system at a very early stage of the development process. Design errors could be detected; design alternatives could be intuitively tested without the need of a real prototype.

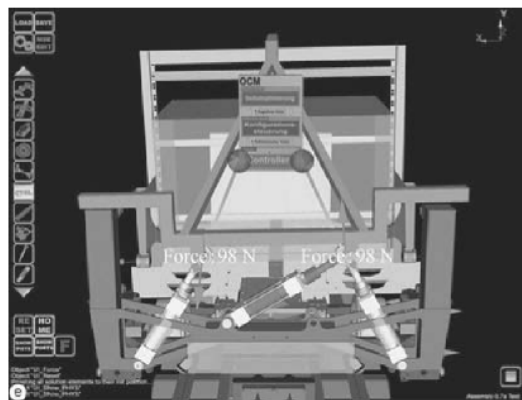


Fig. 8 Visual representation for simulation and analysis

3.2 Virtual Prototyping of Headlight Systems

Modern automobiles contain more and more mechatronic components to support the task of driving. Such mechatronic components are, e.g., an anti-lock braking system (ABS) and an electronic stability program (ESP) to support driving safety, or advanced front lighting system (AFS) to enhance the lighting capabilities of a vehicle on a winding road.

Dynamic bending lights typically use the steering wheel angle or a gyro sensor to calculate the swivelling angle of the headlights to enhance vehicle lighting on a bending road. These systems provide a headlight that “follows” the course of a winding road and are referred to as advanced front lighting systems (AFS) (Fig. 9). The main disadvantage of such systems is that the vehicle lighting heads into a curve too late.

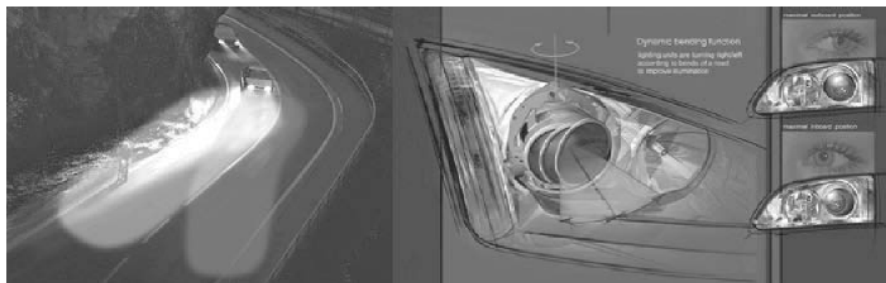


Fig. 9 Advanced Front Lighting (AFS) (Source: Visteon)

Therefore, our project partner Visteon, a global supplier to the automotive industry, has developed a predictive advanced front lighting system (P-AFS) [5]. P-AFS uses GPS-data to locate the vehicle’s position plus digital map data to predict the curvature of the road in front of the vehicle. Based on this, P-AFS predicts the road scenario and turns the front headlights accordingly. That way, the headlights follow the road’s curvature and optimally illuminate the road in front of the vehicle.

In design of headlight systems, we need to do numerous tests for the different headlight. In order to reduce the time and cost spent on the tests, we developed Virtual Night Drive (VND) (Fig. 10), a PC-based night drive simulator [6]. VND utilizes the Shader technology of latest generation graphics systems to visualize the complex lighting characteristics of modern automotive headlight systems in high detail and in real-time during a simulated night drive [7]. The user of VND drives a simulated vehicle on a virtual test track. Different headlight systems are supported and the user can choose between various headlamp models and display modes.

The motion of the vehicle, resulting from the vehicle dynamics and the user’s driving action, directly influences the lighting direction of the vehicle’s headlights. The effect on the illumination of the road in front of the vehicle is visualized in real-time and can be evaluated on screen immediately.

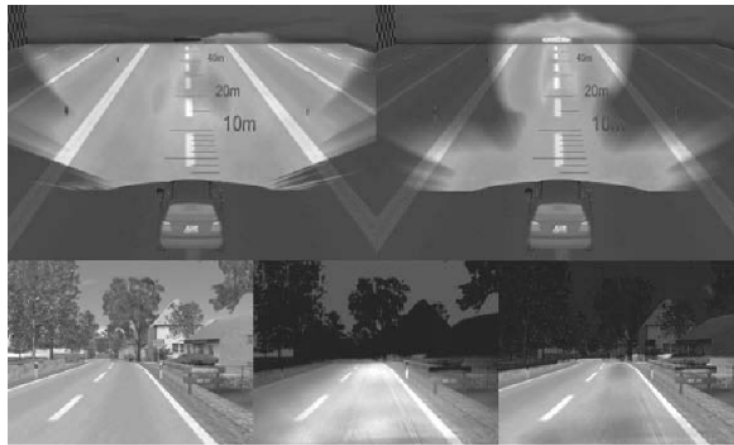


Fig. 10 Real-time visualization in Virtual Night Drive: low beam (upper left), high beam (upper right), and display of day light, night time, and false color modes (bottom half)

The system can be scaled from a single-PC version with one screen and simple steering wheel up to several linked PCs firing a multi-channel stereo display system. Each PC drives one channel of the projection system in mono or stereo mode. A complete cockpit of a small vehicle (Smart-for-two by Daimler Chrysler) is connected to the system as the IO-interface to the user (see Fig. 11). Several test-setups are installed on site at project partners both from academia and industry.



Fig. 11 Flexible presentation setup for VND: scaling VND from simple desktop-version to full-scale simulator application

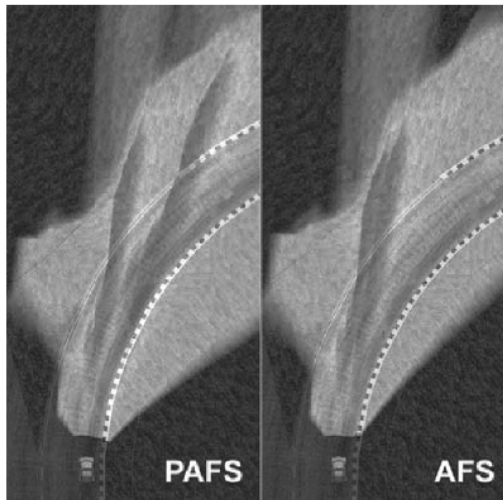


Fig. 12 Evaluation of PAFS within VND: bird's eye view shows early swiveling of PAFS (left) and conventional AFS (right)

The prototypic integration into our VND-system simulates both predictive and conventional AFS and facilitates a direct comparison of both systems on screen. To illustrate the functional behavior of both systems more clearly, different viewing position can be chosen interactively, while the system is running. Bird's eye view (Fig. 12) reveals the different timings between P-AFS and conventional AFS when turning the headlights to follow the oncoming bend of the road.

3.3 AR in Vehicle Prototyping

Currently, the ergonomic design becomes an essential factor in judging the success of a car model. In particular, the increasing features for security, comfort and communication require conclusive view and control concepts to reduce the stress for the driver. For a long time, real prototypes have been used in the conceptual design of vehicles. Physical, 1:1-scale mock-ups of a driver's cabin with seats and dashboard allow the engineer to evaluate the interior design (Fig. 13 left). However, to build or modify such real prototypes is time consuming and expensive. In addition, commercial CAx software based on 3D-modells of the car and the driver facilitate a basic analysis of the ergonomic design (Fig. 13 right). They allow e.g. the calculation of the passenger's leg room or his field of view. Besides those computational methods, the quality of vehicle ergonomics is heavily influenced by perception of the real passenger. For this, the physical mock-up is still in use.

Currently, the ergonomic design becomes an essential factor in

In recent years, Volkswagen Commercial Vehicles developed a VR-based virtual

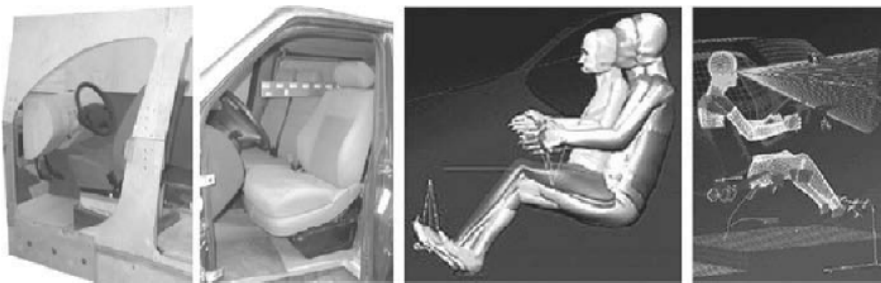


Fig. 13 Physical Mock-Up of the driver's cabin in 1:1-scale (left) and Analysis of vehicle ergonomics using 3D human models (right, Source: Volkswagen Commercial Vehicles AG)

prototyping system (Fig. 14). The test person sits in a real driving seat and wears a head mounted display. The person feels itself located in a virtual car, in which the person can steer through a virtual city. This allows an intuitive analysis of visibility conditions concerning e.g. the instrument panel, traffic lights or pedestrians.



Fig. 14 VR driving seat (Source: Volkswagen Commercial Vehicles AG)

Compared to the physical models, this system has been a step ahead. But the realistic representation of the virtual environment needs a lot of effort, especially the simulation of the vehicle dynamics cost a lot of modeling time and computational power.

In a joint research project together with Volkswagen Commercial Vehicles we thought about a new approach: Why not combine a real car with virtual car components to get best of both worlds? Therefore, we developed an AR-based mobile test bench for vehicle ergonomics [8].

The basic platform of the mobile test bench is a real car without dashboard, rear seats, columns and ceiling. The driver is seated in his normal driving position with a HMD on. The HMD has a pair of integrated video cameras and LC-displays. The cameras grab the driver's view on the real world. The position and viewing direction of the driver are captured by a high precision optical tracking system (Fig. 15 left). While driving the mobile test bench, the driver sees the real car and test track augmented with virtual car components (Fig. 15 right).



Fig. 15 Mobile AR test bench with tracking system for position detection (left) and virtual car components in driver's field of view (right)

This allows an intuitive, fast check of the visibility condition inside and outside the car. Design variants could be easily tested by a simply exchange of the virtual components. In comparison to former approaches, the mobile test bench is much more flexible and less time consuming. The benefit has been approved by several case studies together with engineers from Volkswagen. Now, the project has been extended to integrate further functions like high quality rendering or interactive manipulation of the virtual components. Within the next months, Volkswagen intends to apply the mobile AR test bench in a new car development project.

4 From Mechatronics to Self-Optimization

The term “mechatronics” stands for the symbiotic cooperation of mechanics, electronics, control engineering and software engineering. That opens fascinating perspectives for the development of future mechanical engineering products. The variety of mechatronic systems can be expressed by two categories:

- The first category is based on the spatial integration of mechanics and electronics. The aim is to reach a high density of mechanical and electronic functions within available space. The focus of this class lies on assembly and connecting technologies e.g. Molded Interconnect Devices (MID).
- The second category deals with the controlled movements of multi-body systems. Here sensors collect information about the environment and the system itself. The control system utilizes this information to derive appropriate reactions. These reactions are then executed by the system’s actuators. That is what we call the basic control loop of a mechatronic system.

A representative example for the second category of mechatronic systems is the project “RailCab”. Here the control technology is in the focus of attention. RailCab is an innovative rail system that has been realized as a test facility on a scale of 1:2.5 at the University of Paderborn (www.railcab.de). The core of the system comprises autonomous vehicles (shuttles) for transporting passengers and goods according to individual demands rather than a timetable (Fig. 16). They act proactively, e.g. they form convoys to increase capacity utilization and reduce energy consumption. The shuttles are driven by an electromagnetic linear drive. The main part of the shuttle’s technics is located in its flat floor pan to which the different cabins for passengers and cargo are attached. Figure 16 shows the shuttle itself and the system’s capability to form convoys automatically.

Usually mechatronic systems of this category handle more than one control task, which have to be coordinated. To cope with this complexity our colleague J. Lückel has introduced a hierarchical structure of three levels [9] (Fig. 17).

The basis of this is provided by so called “mechatronic function modules”, consisting of a basic mechanical structure, sensors, actuators and a local information processor containing the controller.

A combination of mechatronic function modules, coupled by information

Demand- and not Schedule-Driven
Autonomous Vehicles (RailCabs) for
Passenger and Cargo

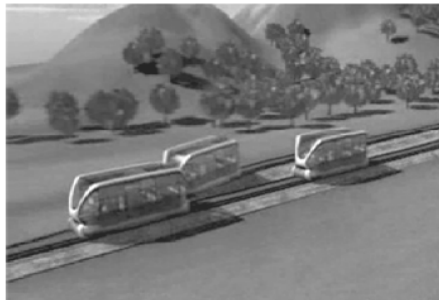
Standardized Vehicles that
can be Customized Individually



Passenger RailCab



Cargo RailCab



Convoy Formation



Comfort Version



Local Traffic Version

Fig. 16 RailCabs of the project “Neue Bahntechnik Paderborn/RailCab”

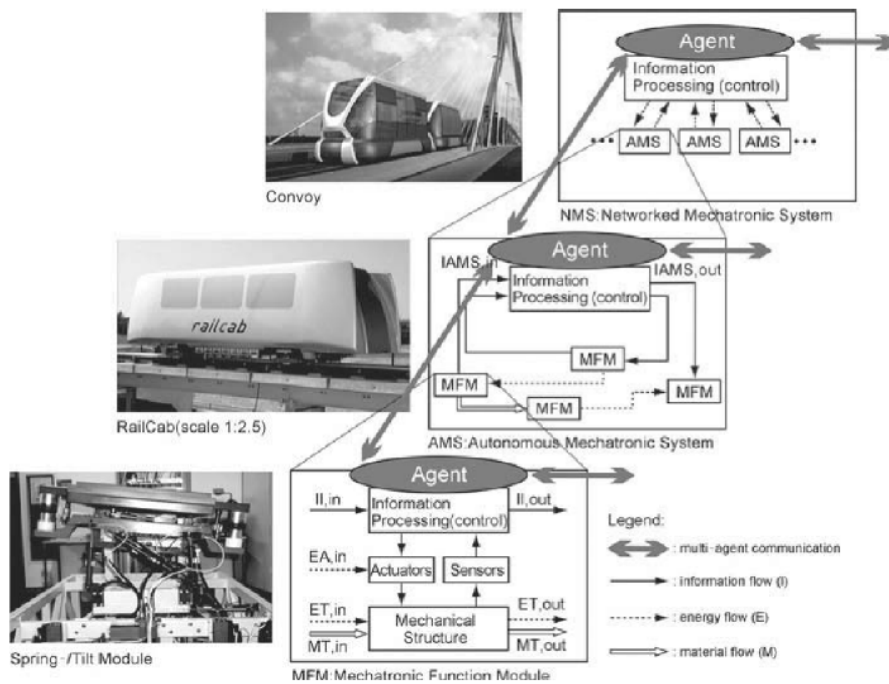


Fig. 17 Hierarchical structure of a self-optimizing system

technology and mechanical elements, constitutes an “autonomous mechatronic system”. Such systems also possess a controller, which deals with higher-level tasks such as monitoring, fault diagnosis and maintenance decisions as well as generating parameters for the subordinated information processing systems of the mechatronic function modules.

Similarly, a number of autonomous mechatronic systems constitute a so called “networked mechatronic system”, simply by coupling the associated autonomous mechatronic systems via information processing.

In the context of vehicle technology, a spring and tilt module would be a mechatronic function module, the shuttle would be an autonomous mechatronic system, and a convoy would be a networked mechatronic system. Because of the high complexity and the participation of different domains the development of mechatronic systems is still a challenge, let alone of self-optimizing systems.

We use the term “self-optimization of a technical system” to describe the endogenous adaptation of the system’s objectives to changing environmental conditions and the resultant autonomous adaptation of its behavior. Self-optimization facilitates systems with inherent “intelligence” that are able to take action and react autonomously and flexibly to changing operating conditions.

Reverting to the hierarchical structure of a mechatronic system (Fig. 17) on each level the controllers are enhanced by the functionality of self-optimization. Thus, the previously mentioned modules and systems (that means mechatronic function modules, autonomous mechatronic systems and networked mechatronic systems) receive an inherent partial intelligence.

The key aspects and the mode of operation of a self-optimizing system are depicted in Fig. 18. Using the influences as a basis, the self-optimizing system determines the internal objectives that have to be pursued actively. These internal objectives are based on external ones, whereas those are set from the outside, e.g. by the user or other systems, and also on inherent objectives that reflect the design purpose of the system. Inherent objectives of a driving module can be for example: ensuring the driving functions and a high efficiency. When we subsequently talk about objectives, we refer to the internal ones, because those are part of the optimization. Low energy demand, high travelling comfort and low noise emission belong to internal objectives of a shuttle. The adaptation of objectives means, for instance, that the relative weighting of the objectives is modified, new objectives are added or existing objectives are discarded and no longer pursued.

Thus, the adaptation of the objectives leads to an adaptation of the system behavior. The behavior’s adaptation is achieved by an adaptation of the parameters and, if necessary, of the structure itself. An adaptation of the parameters means an adaptation of the system’s parameters, e.g. the adaptation of a controller parameter.

Adapting the structure, concerns the arrangement and relations of the system’s elements. We differentiate between reconfiguration and compositional adaptation. It is reconfiguration, if the relations of a fixed quantity of elements are changed. Compositional adaptation means the integration of new elements in the already existing

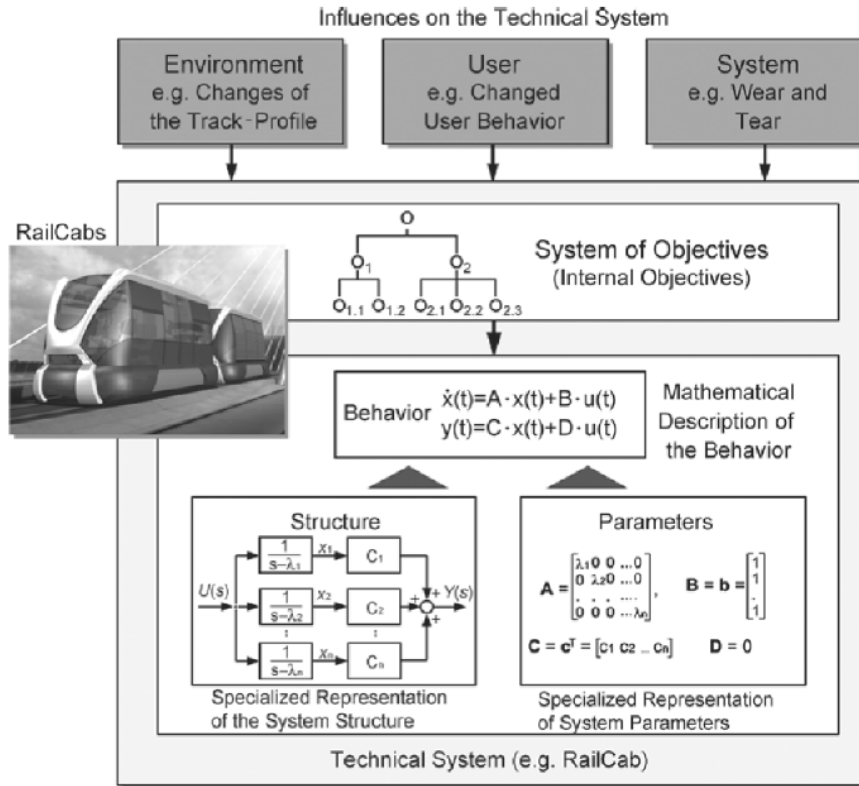


Fig. 18 Key aspects of a self-optimizing system

structure or the subtraction of elements from the structure. Self-optimization takes place as a process that consists of the three following actions, called the **Self-Optimization Process**:

- (1) **Analyzing the current situation:** The regarded current situation includes the current state of the system as well as all observations of the environment that have been carried out. Observations can also be made indirectly by communication with other systems. Furthermore, a system's state contains possible previous observations that are saved. One basic aspect of this first step is the analysis of the fulfillment of the objectives.
- (2) **Determining the system's objectives:** The system's objectives can be extracted from choice, adjustment and generation. By choice we understand the selection of one alternative out of predetermined, discrete, finite quantity of possible objectives; whereas the adjustment of objectives means the gradual modification of existing objectives respectively of their relative weighting. We talk about generation, if new objectives are being created that are independent from the existing ones.
- (3) **Adapting the system's behavior:** The changed system of objectives demands

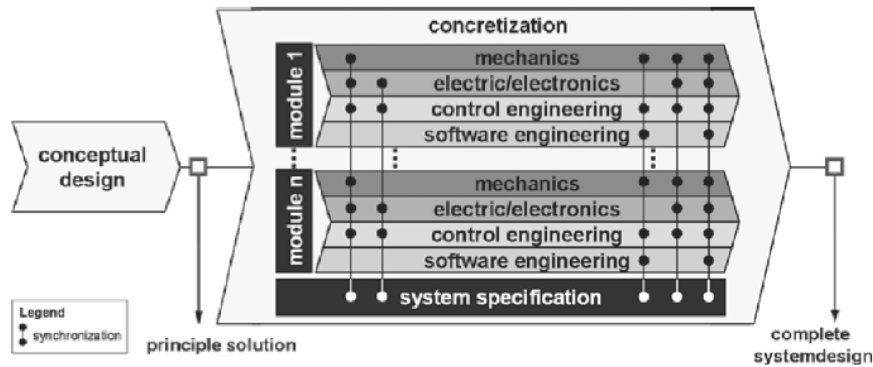
an adaptation of the behavior of the system. As mentioned before this can be realized by adapting the parameters and, if required, by adapting the structure of the system. This action finally closes the loop of the self-optimization by adapting the system's behavior.

The self-optimizing process leads, according to changing influences, to a new state. Thus, a state transition takes place. The self-optimizing process describes the system's adaptive behavior. This can occur on every hierarchy level of a self-optimizing system shown in Fig. 17. The realization of complex, mechatronic systems with inherent partial intelligence requires an adequate concept of structure as well as architecture for the information processing. To make this possible, a new concept has been developed: **Operator-Controller-Module (OCM)** [10]. From an information processing point of view, it corresponds to an agent.

5 Design of Advanced Mechatronic Systems

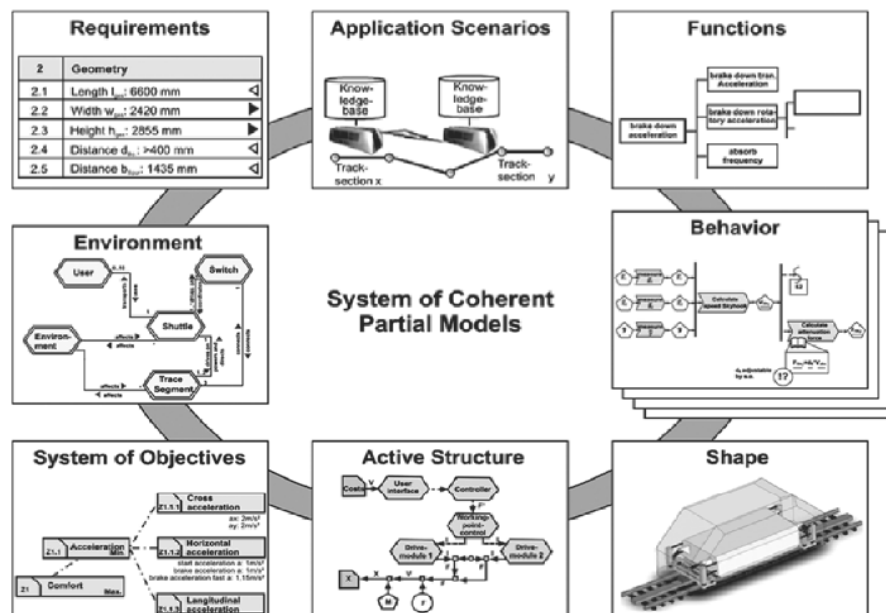
The development of advanced mechatronic systems is still a challenge. The established design methodologies, i.e. the engineering design by Pahl/ Beitz [11] or the VDI Guideline 2206 [12], lay the foundation to meet these challenges. Nevertheless, these methodologies need to be fundamentally extended and added by domain-spanning methods and tools to handle the complexity of the development. This especially applies to the early development phase "conceptual design".

On the highest degree of abstraction, the development process of advanced mechatronic systems can be subdivided into the domain-spanning conceptual design and the domain-specific "concretization" (see Fig. 19). Within the conceptual design, the basic structure and the operation mode of the system are defined. Thus, the conceptual design has to include the decomposition of the system into modules. The decomposition results into a development-oriented product structure, which integrates the two basic and mostly contradictory views of shape- and function-oriented structure. All results of the conceptual design are specified in the so-called "principle solution". A set of specification techniques, in order to describe the principle solution of advanced mechatronic systems, has been developed. By using this specification technique, the system that is to be developed will be described in a holistic, domain-spanning way. The description of the principle solution provides all relevant information for the structuring of the system and forms the basis for the communication and cooperation of the developers from different domains. Based upon the principle solution the subsequent domain-specific "concretization" is planned and realized. The term "concretization" describes the domain-specific design of a technical system, based on the principle solution. The aim of the concretization is the complete description of the system by using the construction structure and the component structure. In so doing, all defined modules are developed in parallel, and each module is developed in parallel in the participating domains (Fig. 19).



5.1 Domain-Spanning Specification of the Principle Solution

Right from the start of our work, it became clear that a comprehensive description of the principle solution of a highly complex system needs to be divided into aspects. Those aspects are, according to Fig. 20: requirements, environment, system of objectives, application scenarios, functions, active structure, shape and behavior. The behavior consists of a whole group because there are different kinds of behavior, e.g. the logic behavior, the dynamic behavior of multi-body systems, the cooperative behavior of



system components etc. The mentioned aspects are mapped on computer by partial models. The principle solution consists of a coherent system of partial models because the aspects are in relationship with each other and ought to form a coherent system. It is necessary to work alternately on the aspects and the according partial models although there is a certain order.

The description of the environment, the application scenarios and requirement serve as the starting point. They are usually followed by the system of objectives, the function hierarchy and the active structure. The active structure represents the core of the principle solution in conventional mechanical engineering. The modeling of states and the state transitions and also the impacts on the active structure play a decisive role in the specification of a self-optimizing system. This kind of modeling takes place within the group of behavior models. The following subchapters explain the partial models, the relationships between the partial models and the specific characteristic of the specification of self-optimizing systems.

The described partial models are not compiled in a predetermined sequence, but in a close interplay. Normally, you start with the partial models requirements, environment, application scenarios and system of objectives.

The core of the system's self-optimization is the self-optimization process and its effects on the system. As mentioned before, the process of self-optimization compares to a state transition of the system. Thereby the system is transformed from one configuration to another, and it changes its behaviour. For the entire description of the self-optimization process, the partial model active structure and two types of the partial model behavior are necessary — please remember there are quite a number of partial models representing behaviour. The two types, that are relevant here, are behavior — states and behavior — activities.

The **partial model behavior — states** describes all relevant states of the system and all events, which could cause a state transition.

The **partial model behavior — activities** specifies the cross-domain behavior of the system. The behavior is described by operation and adaptation processes. **Operation processes** characterize the activities of a system in one specific state: These are for example the acquisition of information about the environment, the derivation of adequate control interventions, and the controlling itself. **Adaptation processes** specify the activities, which are necessary for a state transition. This is the self-optimization process.

The **partial model active structure** comprises all relevant configurations of system elements (controllers, sensors, actuators) and their interrelationships. These interrelationships express flows of energy, information and — if needed — material. In each state, one definite configuration of system elements is activated. State transitions are often realized by additional system elements.

These three partial models are strongly interdependent: To each state of the system in the partial model behavior — states an operation process in the partial model behavior — activities is assigned, as well as a configuration of system elements in the partial model active structure. And to each event in the partial model behavior — states

an adaptation process in the partial model behavior — activities and realizing system elements in the active structure are associated.

5.2 Managing the Development Process

Communication and cooperation between all engineers involved in the development process are essential for a successful and efficient development of advanced mechatronic systems. Within the conceptual design phase, domain-spanning development tasks and their results have to be visualized, so that developers can further elaborate the system design in a cooperative way. In the concretization phase, developers work independently from others on modules in different domains. Their specific development tasks need to be synchronized with those of other domains or modules. Therefore, domain- and module-spanning coordination processes need to be defined and communicated between developers by means of appropriate visual tools. We identified the following user tasks that need to be supported by a visual presentation tool:

Overview and details: Developers have to synchronize their work with other activities in the development process and need to identify which process steps relate to others and which can be refined and executed independently. This requires a visual overview of the complete development process as well as a focus on specific and detailed information of individual development tasks and their results.

Search, filter and results: The large number of process steps requires an efficient search mechanism. Arbitrary types of process elements should be selectable by filter operations and results should be represented and accessed efficiently.

Interactive navigation and presentation: Communication between developers is best realized by means of an interactive visualization of the complete system model. Users must be able to efficiently navigate through the complex model and present selected elements at an arbitrary level of detail. Moreover, it should be possible to present a sequence of user-defined views of the system model, for example to discuss a specific workflow.

The aim of the visual presentation tool is support the design team in managing the complex development process of advanced mechatronic systems. With such a tool, we break new grounds in process visualization by applying methods and techniques from classical design review not only to the results of individual development phases, but rather to the entire development process itself.

We chose a complex application example by selecting the complete design model of the RailCab prototype, in order to demonstrate the productivity of the visual presentation tool. The development process of the RailCab comprises of 850 development steps including about 900 development objects, e.g. kinematic models, dynamic models, controller models, state charts, list of requirements, test results etc. During the concretization phase seven modules are developed in detail. To facilitate this process, we developed a zoomable user interface that visualizes the complete development process in high detail on a high resolution projection system [13] (Fig. 21).

Figure 22 illustrates, how the visual presentation tool could look like. A zoomable view of the entire development process supports a quick navigation through the complete system model. The user can efficiently search the complete system model, navigate to any process steps of the entire process model, switch between arbitrary level of detail and present results of individual process steps, like for instance a partial model of the principle solution.

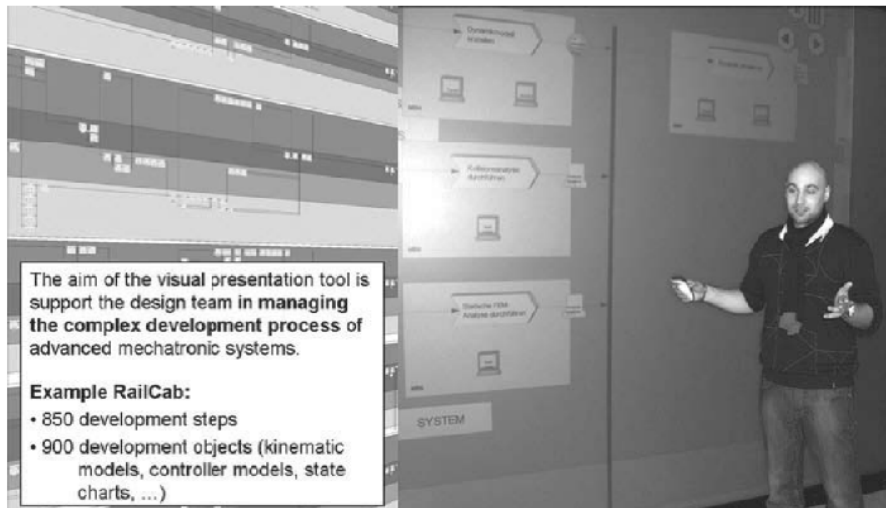


Fig. 21 User in front of a presentation wall (right) and overview of complete development process of the RailCab (left)

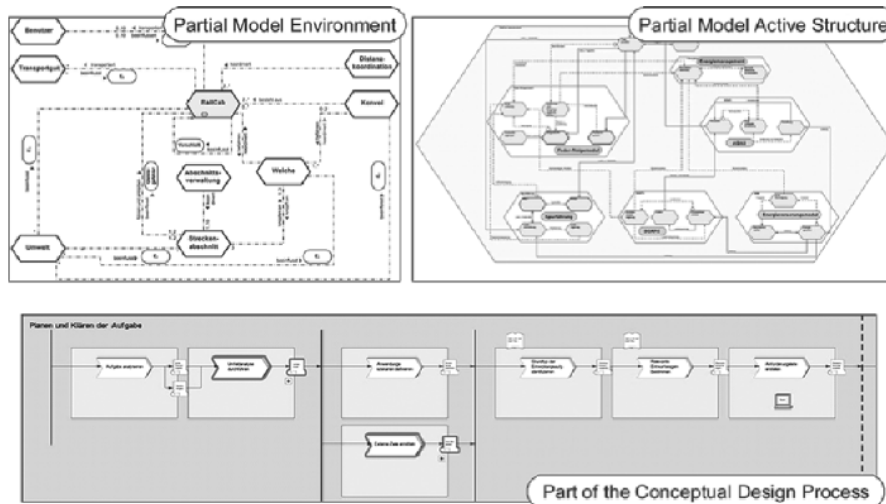


Fig. 22 Visual presentation tool for managing the development process. Partial model environment (top left), overview of the active structure (top right) and extract from the development process of the RailCab (lower half)

6 AR- and VR-based Testing of Advanced Mechatronic Systems

For the development of an advanced mechatronic system like the RailCab, we first apply the specification technique, which is introduced in the previous section, to define a principle solution of the RailCab. Based on the principle solution, we then develop numerous virtual prototypes and test beds for real prototypes, in order to analyze and evaluate the behavior of selected RailCab components and modules. However, the analysis and evaluation of the RailCab and its components can get complex and time-consuming, due to two main reasons:

- The RailCab can show a quite complex behavior, e.g. when a RailCab joins or leaves a convoy. These processes elapse fast and involve numerous variables and data, making it difficult for the engineer to comprehend the course of action and to maintain an overview of the whole procedure.
- Some prototypes of the RailCab components, e.g. the test bed for RailCab's undercarriage, operate swiftly and their parts mostly move by merely a few millimeters. For the engineer, such tiny motion is hard to perceive.

For these reasons, we apply two approaches, one AR- and one VR-based, to facilitate the analysis and evaluation of selected prototypes and their test beds:

- For the AR-based approach, we combine variables and test data from the prototype and its test bed and utilize the technology AR to augment the engineers view on the system by superimposing relevant information into the user's view.
- For the VR-based approach, we apply the technology VR to visualize missing components, like e.g. the test track, in order to simulate, how well the undercarriage prototype performs on alternative test tracks, other than our real test track on the campus in Paderborn.

In the following subsections, we describe these two examples, which demonstrate our AR- and VR-based approaches to facilitate the analysis and evaluation procedures for two prototypes of the RailCab.

6.1 AR-based Visual Analysis of the Convoy Behavior

In the first example, we apply the technology AR to facilitate the analysis of the convoy building process of several RailCabs on the test track on the university campus in Paderborn. As we only have one real RailCab prototype available for driving on the test track, we need to simulate additional virtual RailCabs to be able to simulate convoy building processes. Based on AR-technology, we project the virtual RailCabs into a video of the real RailCab which is driving over test track. The AR-application combines and visualizes all relevant information directly on screen, so that the engineer can easily observe the convoy building process on one monitor. Furthermore, the AR-application allows to check, how well an alternative version of the control software cooperates with previous versions during a convoy building process.

To evaluate the concept of convoy driving and the convoy controller under realistic

conditions, a hardware-in-the-loop-system has been developed [14]. In this HIL-system, a real RailCab and four simulated RailCabs have been combined to a convoy. The principle is shown in Fig. 23. The real RailCab is driving over the test track at the university campus and serves as the convoy leader. The convoy control unit inside this vehicle manages the entire convoy. The concept of autonomously driving convoys requires the RailCabs to drive consecutively in a distance of less than one meter without any mechanical coupling [15–17]. To realize this, the RailCabs must communicate among each other. Therefore, the leading RailCab transmits its position and leading speed to the four simulated vehicles, which use this for guidance. All sensors, actuators and the mechanic parts of the virtual RailCabs are simulated. To avoid collisions between the RailCabs inside the convoy, each vehicle is distance-controlled by a subordinated speed controller. The distance is regulated relative to the position of the RailCab driving ahead.

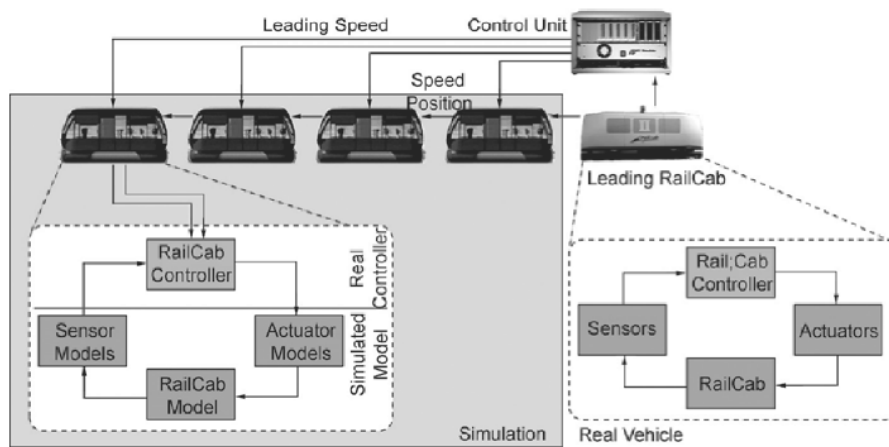


Fig. 23 Principle of the hardware-in-the-loop test of the RailCab convoy

The limitation of the HIL-simulation is that the simulated RailCabs are not visible on the test track. Their movement can only be illustrated by a timeplot. However, when using the HIL-based test bed, it is difficult to test different control strategies and parameter settings interactively during online experiments. Usually, the engineer has to conduct an experiment at the test track and collect and save the data for later analysis in the lab. Changing the strategies or parameter settings interactively during an ongoing experiment is limited and complicated. An AR-based visualization of the simulated RailCabs and corresponding data, that explains their behavior, enables an interactive testing of the HIL-simulation.

For this reason, we have developed an AR-application to visualize the simulated RailCabs and additional parameters on the real test track. We facilitate a visual analysis of the convoy behavior at the HIL-based test bed. This enables the engineers to understand the behavior of the HIL-test during an online-experiment. Main question

for analysis is: Is the convoy able to drive without any collision, using different control strategies and parameters? For this, we visualize the shape of the simulated RailCabs and their motion on the real test track, as well as abstract control data such as position and speed of each RailCab, the distance between two RailCabs, braking distance and leading speed. In the following, we present four subjects for analysis, in order to illustrate how the AR-based visualization facilitates the testing of the HIL-based test bed and the evaluation of the convoy behavior.

Convoys: One important question for an energy-efficient operation of the RailCabs is, how well a convoy can be established and maintained. Figure 24 shows a convoy consisting of a real RailCab being followed by three virtual RailCabs. Seeing the simulated RailCabs superimposed on the test track help the engineers to recognize, whether the RailCabs drive in a convoy or whether a RailCab loses contact to a convoy. In Fig. 24 the last RailCab has lost contact to the convoy and the gap between the last RailCab and the preceding convoy is too large to take advantage of the slipstream.

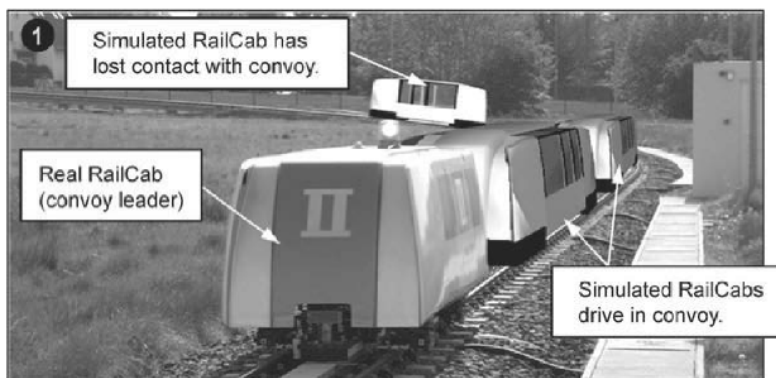


Fig. 24 AR-visualization of a RailCab losing contact with the convoy

Collisions can occur, when RailCabs try to minimize their distance to preceding RailCabs in a convoy, in order to reduce air resistance and save energy. Figure 25 illustrates the AR-based analysis of a collision between two RailCabs. If two RailCabs collide, their shapes are highlighted in red. If the distance between two RailCabs falls below a specified limit, a collision warning indicates an upcoming possible collision. In the visualization, this is indicated by a flashing RailCab.

The **control quality** is an important indicator that helps to avoid critical conditions while driving in a convoy. In the AR-based application, we visualize such abstract quantities by plausible and easy-to-understand color-codes. Figure 26 shows a visualization of the control quality of each RailCab using such a color-code.

Here, RailCabs are highlighted in a color range spanning from green over yellow to red. The color shades indicate, how well the controller in each RailCab of the convoy is able to follow any reference values, the superordinated convoy controller in the leading RailCab, which functions as the convoy leader, communicates to all convoy members.

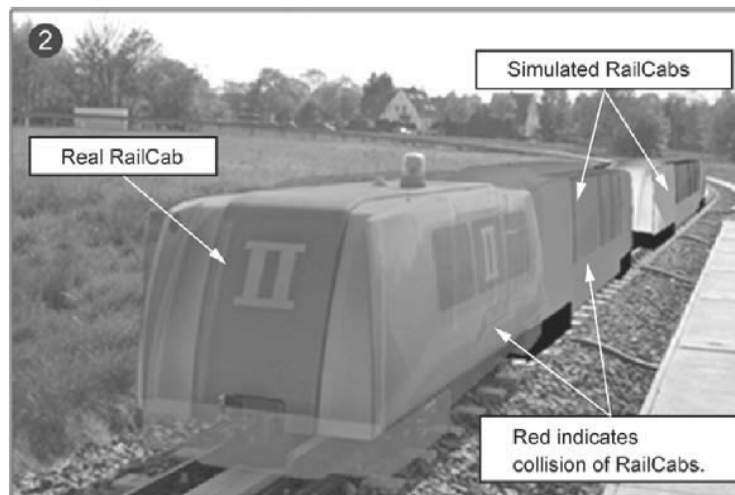


Fig. 25 AR-based analysis of a collision occurring in a convoy

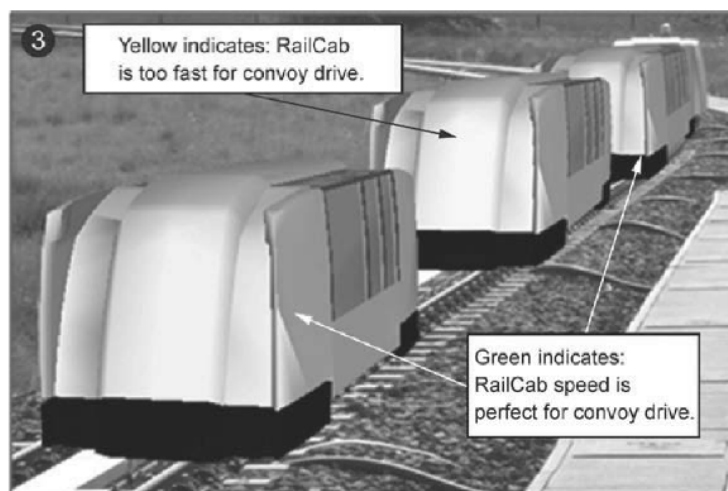


Fig. 26 Supervision of control quality

Braking distance: In case of a system or component failure, one of the crucial parameters for the convoy controller is the braking distance of each RailCab in the convoy. The braking distance directly influences the convoy behavior and helps to prevent rear-end collisions for a save operation of the convoy. For a save convoy drive, the braking distance of a following RailCab in a convoy must, at any time, be smaller than the braking distance of the preceding RailCab. To visualize the braking distance, a red bar is superimposed on the track (see Fig. 27). The highlighted track section indicates the estimated braking distance of the RailCab. The vertical red bar at the end of the highlighted track section indicates the expected final position at which the RailCab will come to a complete stop.



Fig. 27 AR-based visual analysis of the braking distance and the final position of a RailCab

6.2 Realization of the Augmented Reality System

Figure 28 shows the HIL-test bed setup on the left and the setup of the AR-system on the right. A communication server is integrated between both components. The lower half of Fig. 28 shows the operator's control desk for observing the test track.

The HIL-test bed is composed of a real RailCab and a real-time prototyping system, produced by dSpace. A simulation of four real RailCabs is running on the real-time system under hard real-time conditions. The real RailCab on the test track is the leader of the simulated convoy. This RailCab communicates with the simulation system by an internal, proprietary bus system.

The simulation system sends position and speed data of all RailCabs to the communication server. The server is used as an interface between the HIL-test bed, running under hard real-time conditions, and the AR-system, running under soft real-time conditions. Every 10 ms the HIL-test bed sends data of the RailCabs to the communication server. Since the AR-system just renders a new image every 40 ms, it gets each fourth data package from the communication server.

The AR-system is built up from a PC with a Pentium 4, 2.8 GHz Dual Core processor, a NVIDIA 8800 GTX graphics board and 2 GB RAM. The output device for the video is a HDTV monitor with a wide screen ratio of 16:9. For recording the video of the test track, a Sony HVR-V1E HDV video camera is used. The monitor and the camera both have a resolution of 1980×1080 pixels. We need the high resolution to be able to monitor not only RailCabs that pass by near the camera, but also those RailCabs that drive on the far side of the test track.

The AR-software application was implemented in C++ and the 3D graphics toolkit OpenSceneGraph, which offers various functions to render animated 3D objects. The video image is rendered as a 2D image in the background with the fragment shader of the graphics board. To demonstrate the simulated RailCabs in the AR-application, CAD models of the real RailCab were integrated into the application.

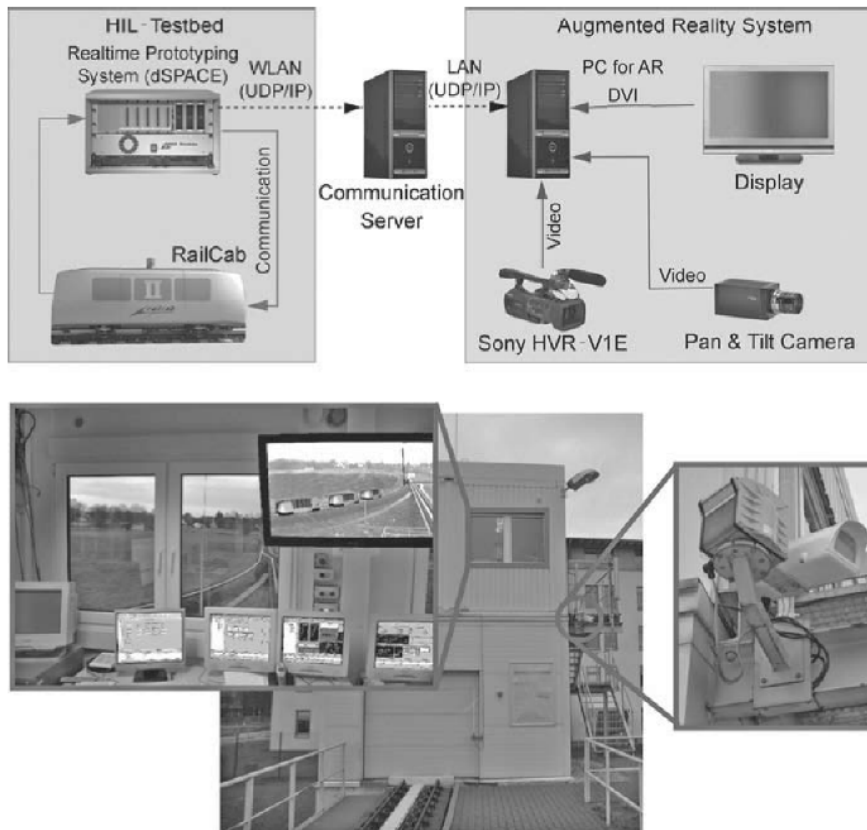


Fig. 28 Setup of the augmented reality system at the test track

6.3 VR-based Visual Analysis of the RailCab Undercarriage

The second example describes the use of the technology VR, in order to facilitate a targeted testing of the RailCab undercarriage components as well as for the controller development. A test bed for the RailCab undercarriage in scale 1:2.5 has been developed at University of Paderborn. The RailCab undercarriage actively steers over passive switches, enabling the RailCab to choose its direction at a switch.

As the wheels are not forced along the tracks, like with conventional trains, guidance along a track and steering over a switch require frictional forces. These are particularly important on wet or icy tracks. This led us to developing a camber-variable undercarriage, which actively controls the camber of the wheels. The aim is to uncouple guidance and steering forces from the frictional forces between wheel and track. In addition, the undercarriage actively steers each wheel individually, in order to reduce torque and frictional forces considerably when steering the RailCab over a switch.

The principle is shown in Fig. 29. On the left, Figure 29 shows a profile of a track

and a wheel with varying camber angles and without changed camber angle. The track has a spherical cross cut profile, so that the wheels can support themselves on it while cambering.

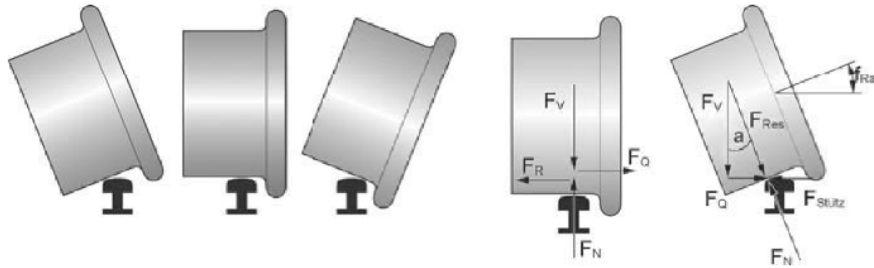


Fig. 29 Varying camber angles of a wheel (left) and distribution of forces at a wheel in normal position and an inclined wheel (right)

On the right of Fig. 29 it is shown, how the forces are distributed at a wheel in normal position and an inclined wheel. If the wheel is standing up straight, the RailCab is kept by the friction force F_R on the track. The friction force compensates the shear force F_Q , which can e.g. be generated by centrifugal force. When a wheel cambers, it supports itself on the track mechanically. The shear force is compensated by the profile lateral force $F_{Stütz}$ at the contact point of the wheel, but not by the friction anymore.

6.4 Test Bed of the Undercarriage

Figure 30 shows a RailCab undercarriage and its test bed. The test bed of the undercarriage comprises a wheel suspension of the RailCab, which is fixed in a frame. In the center, two actuators, which adjust the camber angle of the wheels on both sides of the undercarriage, are mounted to the frame. Each wheel is connected to another actuator that adjusts the wheel's steering angle. The wheels are positioned on a short piece of a track. Sensors for measuring the distance between wheel and track are mounted to the undercarriage.

For the active steering of the wheels of the undercarriage, a controller computes the camber and steering angles of each wheel in real time and operates the actuators for setting camber and steering angles as well as distance sensors for measuring the position of the wheels on the track.

Since the physical test track is not directly visible from the test bed, it is not possible to simply map the dynamic behavior of the undercarriage prototype on the test bed with the RailCab prototype on the test track. This handicaps a targeted analysis of the prototype's dynamic behavior at the test bed, since data first needs to be collected and evaluated after tests have been finished. In general, the prototype's dynamic behavior can however be evaluated without any specific visualization. But since there is no representation of the test track, which facilitates mapping steering and inclination motion at the test bed to the course of the test track, evaluation is clearly more difficult.

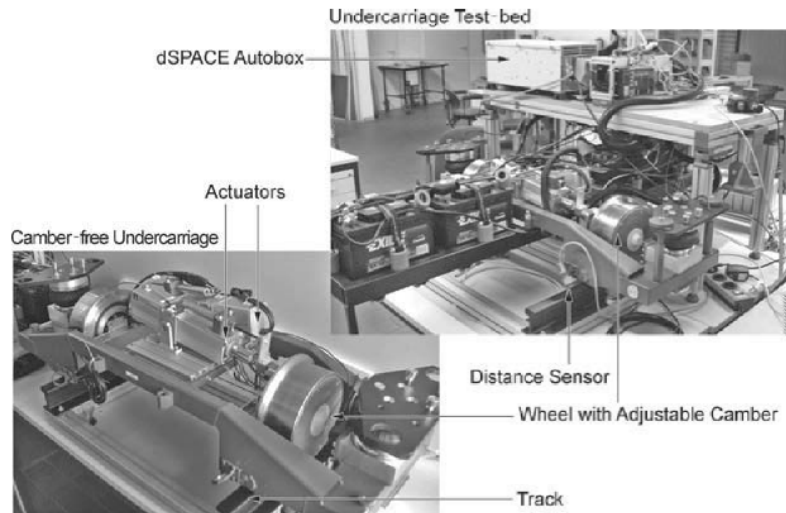


Fig. 30 Camber-variable undercarriage and test bed

6.5 Virtual Reality Application for Visualization

We have developed a virtual reality-based application to facilitate a fast allocation of the camber-variable undercarriage behavior to the course of the test track. The VR-application comprises a visualization of a virtual test track and a virtual model of the camber-variable undercarriage. The application is connected with the undercarriage test bed via a real-time bus. This ensures that steering and inclination angles of the wheels in the test bed correspond with the virtual model of the undercarriage in the VR application. Figure 31 provides an overview of the entire system. The virtual test track serves as a track model for the undercarriage test bed, providing the course of the track, including all slopes, courses of curves, switches, etc., the undercarriage is driving over. The undercarriage adapts the steering and camber angles of its wheels in the test bed according to the course of the virtual test track. While driving along the virtual test track, the VR model of the undercarriage also adjusts the steering and camber angles accordingly and thus demonstrates the dynamic behavior of the prototype in the test bed, as if this would be driving along the virtual test track.

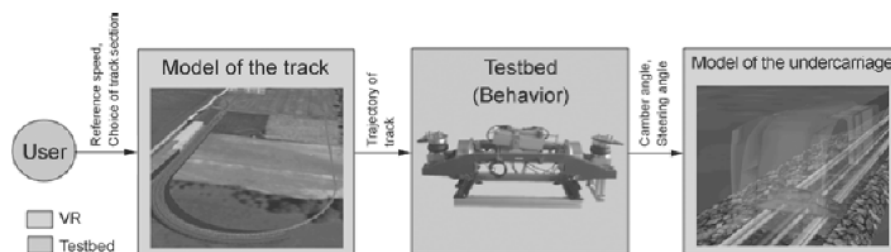


Fig. 31 Overview of the VR application for visualizing the undercarriage on a virtual test track

The VR environment comprises a virtual model of the RailCab test track, which is located on campus at University of Paderborn [15, 18]. Figure 32 right shows an overview of the test track. The track consists of 2 loops, which are connected with each other by 2 switches. The right hand loop has a length of approx. 430 m with a maximum slope of 4,3 %. The left hand loop is about 440 m long with a maximum slope of 3,0 %. While the left hand loop exists only as a virtual model, the right hand loop has a real counterpart on campus at University of Paderborn.

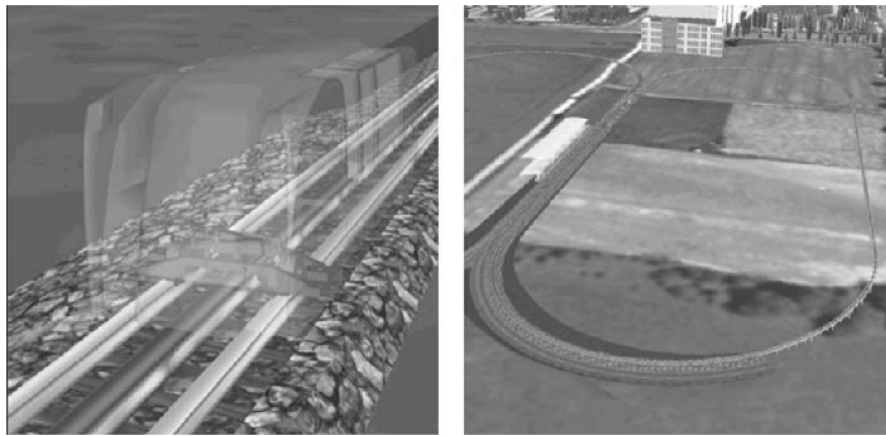


Fig. 32 3D-model of camber-variable undercarriage (left) and virtual test track (right)

On the left, Figure 32 shows the 3D-model of a RailCab with its camber-variable undercarriage on the virtual test track. The RailCab is 3 m long, 1,2 m tall and 1,5 m wide. It is rendered transparently, in order to reveal the camber-variable undercarriage, which is located below the RailCab chassis. The 3d-model of the undercarriage was extracted from 3D-CAD data. All major parts that are essential for demonstrating the functional behavior, such as wheels, actuators for steering and camber angle adjustment, frame, single wheel suspension and sensors, were integrated into the visualization.

For rendering the VR-environment (cp. Fig. 32, right) we use the C++-based scenegraph API OpenSceneGraph, which provides basic elements for creating and efficiently administrating graphical objects for a 3D visualization.

Within the test bed, the undercarriage is activated by data from a simulated test track. The resulting response of the undercarriage can be studied directly at the test bed. As the test track is just simulated and not visible at the test bed, the engineer cannot directly correlate the course of the test track, being the source of the actuation, with the response of the undercarriage, i.e. its' steering and camber angles.

For this reason, we developed a VR-application, which shows a 3D-model of the undercarriage on the virtual test track, in order to facilitate a fast correlation of the dynamic behavior of the camber-variable undercarriage and the corresponding section of the test track the undercarriage is passing over. The VR-application is connected to

the undercarriage test bed and visualizes the steering and camber angles of the wheels based on the current course of the track. This allows to cross-check the motion of the undercarriage relative to the virtual test track and to analyze critical driving situations, e.g. when the RailCab is driving over a switch.

Figure 33 shows an overview of the system. The VR-based visualization is connected with the real time controller of the test bed. That way, the visualized camber and steering angles of the virtual undercarriage correspond to the test bed. The virtual test track provides data of the course of the track (e.g. inclinations, drops, curvature, switches, etc.) based on which the real time controller of the undercarriage computes corresponding camber and steering angles for each wheel. These angles are sent to the actuators in the undercarriage test bed as well as fed back into the VR-simulation for direction comparison. That way, the simulated drive over the virtual test track can be analyzed on the undercarriage test bed.

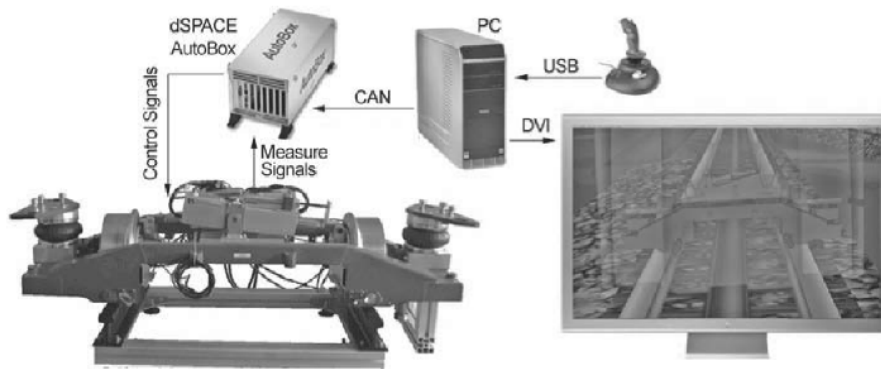


Fig. 33 Overview of the VR-based system for testing the undercarriage on different virtual test tracks

The VR-based testing application provides an interactive steering mode, where the engineer can control the steering angles via a joystick and study the effect of the steering command on the resulting camber angles and the overall behavior of the undercarriage. Also, the speed or the desired driving direction at a switch can be controlled, in order to study the system response.

6.6 Visual Analysis of the Behavior

We have conducted numerous test drives, for which the undercarriage test bed was connected to the VR application for visualizing the simulated test drives along a virtual test track. We performed visual analyses of the test drives, where we focused on examining steering and camber angles and analyzing the dynamical behavior of the undercarriage on the test track and especially during the process of crossing a switch.

Visualization of steering and camber angles: Within the VR-application, we visualize the camber and steering angles of each wheel, in order to facilitate a better

understanding of the behavior of the camber-variable undercarriage. Figure 34 shows two screenshots from the VR-application with a visualization of the camber angles of the wheels, as provided by the real time controller and being set by the actuators. The VR-application shows the undercarriage in a close-up view. That way, during a simulated drive over the test track, the resulting motion of all moving parts within the undercarriage can easily be analyzed by the engineer. Hardly visible objects, like for instance small moving parts or abstract information, are visualized by plausible illustrations.

On the left of Fig. 34, camber angles values are visualized by the green segment within the yellow semicircle which indicates the maximum valid range for the camber angles. Analog to this, steering angles are shown on the right of Fig. 34. Here, the same visualization metaphor is used to illustrate the steering angles with a different maximum valid range and different colors.



Fig. 34 Visualization of camber (left) and steering angles (right) of two wheels in the RailCab undercarriage

Analysis of the undercarriage's dynamical behavior: A fast information acquisition at the physical test bed is crucial for this work. This applies particularly for mechatronical systems that operate under hard real-time conditions and which adapt their behavior within a split second. The aim of the visualization is, to demonstrate the camber-adjusting behavior of the undercarriage during a simulated drive. Since there is no test bed available directly at the test track, it is difficult to identify a correlation between the driving behavior of the undercarriage and the test track.

We describe the analysis of the undercarriage's dynamic behavior in the following example. In this example we want to analyze the algorithms and strategies for driving over a switch. Main question here is, when does the RailCab start to adjust the steering and camber angles of its wheels and when does it turn the wheels back into normal steering position, as well as back to neutral camber angles? This is an important question, since considerable time elapses from the moment, the controller passes the set points to the actuators until the actuators have accomplished adjusting the steering and camber angles in the undercarriage. This needs to be optimized by adjusting the controller parameters.

Analysis of a switch-crossing: Figure 35 shows a RailCab, which is crossing a switch. In Fig. 35 top left, the RailCab is approaching the switch, while the steering angle is still 0° . When the RailCab drives into the switch a steering angle is calculated, which is then set in the test bed setup of the undercarriage and is also

visualized in the VR-application. Due to this, the wheels turn to the left (cp. Fig. 35 top right). As the RailCab enters the switch, the course of the track turns from slight left to slight right. The wheels need to steer along this change of direction and also turn slightly from left to right (cp. Fig. 35 top right and lower left). When the RailCab exits the switch, the tracks turn straight again and the steering angle returns to 0° (see Fig. 35 lower right).

While we develop the control algorithms for the actuators, we still need to optimize the dynamic behavior of the undercarriage. For instance, it may well be possible that, while the RailCab enters a switch, a steering wheel is determined that turns the wheels to the left. Since the RailCab keeps on driving during the process of setting the steering angle, the steering angle is not completely adjusted until the wheels already need to turn back to the right again. At the test bed, the visualization of the test track helps to easily identify and solve this problem. This facilitates a targeted analysis of control algorithms and strategies.

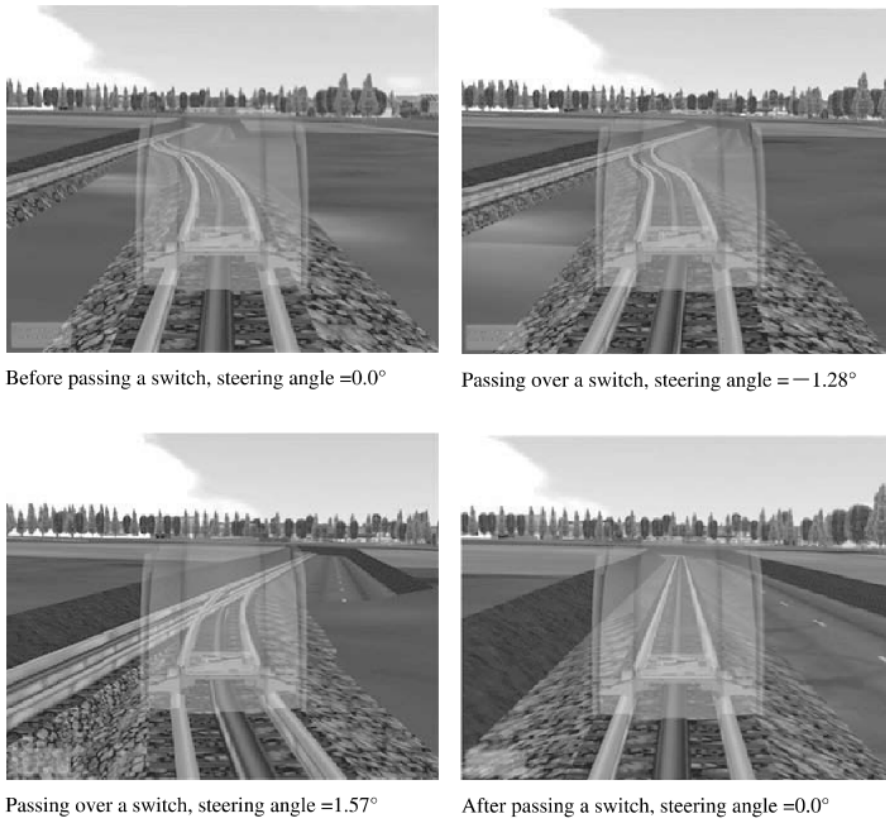


Fig. 35 Analysis of a switch-crossing: Approaching the switch with steering angle 0° (top left), entering the switch with steering angle -1.28° (top right), crossing the switch with steering angle 1.57° (lower left) and exiting the switch with steering angle returning back to 0° (lower right)

7 Conclusion

The development of advanced mechatronic systems is a challenging task which requires know-how of numerous experts from various disciplines. Our new specification technique for the design of such systems facilitates this interdisciplinary process by combining the contribution of each discipline and applying one common specification technique for the conceptual design. We enhance the development of advanced mechatronic systems by integrating methods and techniques from classic design review into a visual presentation tool and applying these methods to the development process itself.

Based on the experience we gained, when applying the AR- and VR-based approaches to some of RailCabs' prototypes and test beds, we can confirm that these approaches facilitate a fast visual analysis and a targeted testing of future advanced mechatronic systems. However, integrating AR- and VR-based approaches into RailCabs' prototypes and test beds is still a time-consuming and complex task, because various simulation models and their interfaces need to be adapted. Thus, our future work focuses on automating the generation of appropriate AR- and VR-based visualizations as well as automatically connecting the visualization components with corresponding simulation models of the prototypes and test beds.

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